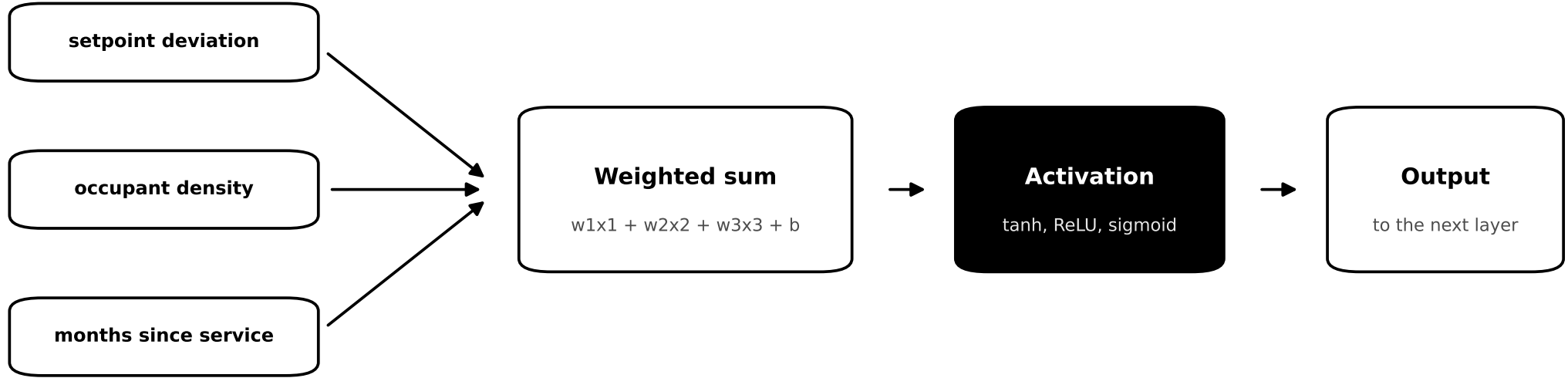


One neuron is a weighted sum followed by an activation

each input carries
its own weight



Without the activation the whole network collapses into a single weighted sum,
so a stack of linear layers can express nothing that one linear layer cannot.